

Article

Bi-Objective Resilient Backbone-Grid Planning via a Three-Stage TER-NSGA-II Approach Considering Pumped-Storage Hub Effects

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Abstract

In the global transition toward low-carbon power systems with high renewable energy penetration, pumped storage has emerged as a strategic cornerstone for modern power grids. However, the collaborative planning of pumped storage and backbone-grids faces critical challenges, including the lack of explicit quantification of the resilience value of pumped storage and the coarse treatment of N-1 connectivity constraints. This paper proposes a bi-objective resilient backbone-grid planning approach that integrates the pumped-storage hub effect, aiming to minimize total life-cycle costs and the system resilience mismatch index. The proposed framework incorporates network connectivity, N-1 connectivity (edge connectivity ≥ 2), and dual-scenario power flow security as rigid constraints. Furthermore, a three-stage constrained evolutionary algorithm TER-NSGA-II is developed. During the N-1 connectivity reinforcement phase, the max-flow min-cut theorem is employed to achieve precise validation and guidance for edge-connectivity enhancement. Case studies on the IEEE 118-bus system, together with extended validation on the IEEE 300-bus system, show that the proposed method can explicitly quantify the resilience value of pumped storage, obtain Pareto solutions that balance economy and resilience under strict edge-connectivity constraints, and demonstrate competitive overall performance in terms of solution-set quality, feasible-domain search stability, and scalability compared with NSGA-II and the more recent NSGA-III/NG benchmark.

Keywords: pumped storage; backbone-grid planning; power system resilience; N-1 connectivity constraint; TER-NSGA-II

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1. Introduction

In the global low-carbon transition of power systems with high renewable energy integration, pumped-storage has emerged as a core strategic resource for enhancing grid resilience, with global plans targeting capacity expansions of tens of gigawatts [1–3]. However, systematic planning challenges remain, including disjointed site selection and transmission-corridor planning, the lack of explicit quantification of the resilience value of pumped storage in black-start scenarios [4], and the coarse handling of hierarchical constraints, including connectivity, N-1 connectivity, and power flow security. Current

planning methods face challenges in balancing computational efficiency with solution quality [5,6]. Therefore, developing a bi-objective resilient backbone-grid planning approach that simultaneously integrates economic performance and system resilience, incorporates precise N-1 connectivity analysis, and fully exploits the pumped storage hub effect is of significant theoretical and engineering importance [7,8].

Current research on pumped-storage and backbone-grid-coordinated planning primarily focuses on economic and technical feasibility, while paying insufficient attention to the resilience-support value of pumped-storage under extreme conditions [9,10]. Existing studies on resilience value modeling predominantly focus on the functional implementation of pumped-storage equipment. Although resilience-support pathways for pumped-storage systems have been discussed from perspectives such as black-start control, emergency output regulation, and rapid load response, quantitative models that explicitly characterize the contribution of pumped-storage units during post-fault system recovery are still lacking. This results in insufficient support for differentiated resilience evaluation of planning schemes and complicates the accurate assessment of the unique resilience value of pumped-storage resources [11–13]. Regarding N-1 connectivity constraints, traditional methods have significant limitations: optimization approaches using penalty functions or slack coefficient methods for N-1 overload expectations are essentially soft constraint designs. These methods cannot strictly satisfy the rigid safety requirement of edge connectivity ≥ 2 , and in complex network topologies, they tend to generate numerous infeasible solutions, consuming excessive computational resources [14–16], thereby reducing planning efficiency. Meanwhile, few studies addressing rigid constraints encounter issues such as exponential computational complexity and convergence difficulties, making them unsuitable for large-scale backbone-grid planning scenarios. In developing evaluation metrics for planning, existing models primarily prioritize economic viability and resource compatibility. While current pumped-storage site selection frameworks incorporate multiple indicators, such as investment costs, output capacity alignment, and environmental benefits, they systematically exclude the resilience dimension. This oversight leads to planning outcomes that fail to adequately address both system resilience under extreme conditions and recovery efficiency [17–21]. The critical gap in this field lies in explicitly quantifying the resilience value of pumped storage, precisely satisfying the stringent N-1 connectivity requirements, and fully integrating resilience assessment into planning models [22].

This study addresses the identified research gaps by investigating both the planning model and intelligent optimization algorithm levels. At the model level, with pumped storage serving as the core power source for system black start, the resilience evaluation system incorporates restoration-distance contribution and capacity-weighted betweenness centrality. A dual-objective “economy-resilience” planning model is constructed, with connectivity, N-1 connectivity (edge connectivity ≥ 2), and dual-scenario power flow security of pumped-storage units as rigid constraints. At the algorithm level, a bi-objective three-stage evolutionary algorithm, TER-NSGA-II, is proposed within the computational intelligence framework. The optimization process is divided into three phases: connectivity construction, N-1 connectivity reinforcement, and safety convergence. During the connectivity reinforcement phase, edge connectivity is precisely validated using the maximum flow-minimum cut theorem. Simulation results from the IEEE 118-bus system, together with extended validation on the IEEE 300-bus system, demonstrate the effectiveness of the proposed method in terms of optimization accuracy, solution-set quality, and engineering applicability.

It should be noted that this study adopts a deterministic modeling framework to represent typical operating scenarios and does not explicitly account for renewable generation variability or load uncertainty. Existing studies have shown that uncertainty can

significantly affect the capacity configuration and operation results of power and integrated energy systems [23,24]. In addition, DC power flow is employed to improve computational efficiency at the planning stage, while voltage magnitude constraints, reactive power effects, and network losses are not explicitly modeled. Therefore, the results of this study are more suitable for candidate backbone-grid screening at the planning stage rather than for detailed operational validation. These limitations are further discussed in the model description and conclusion sections, together with possible directions for future extension.

2. Resilient Backbone-Grid and Pumped-Storage Resilience Value

2.1. The Connotation and Construction Criteria of the Resilient Backbone Grid

In extreme disaster scenarios, a resilient backbone grid is a core network structure designed to ensure continuous power supply to critical loads, formed by optimized transmission channels with topological redundancy and rapid recovery capabilities [25,26]. When pumped-storage systems are extensively integrated as black-start power sources, elevating them to “resilient cores,” their hub locations and bidirectional power support capacities must be incorporated into planning [27].

The resilient backbone-grid framework under study is defined as follows: given fixed pumped-storage connection locations and capacities, the resilient backbone grid is constructed through optimized backbone-channel selection under hard constraints of connectivity, N-1 connectivity (edge connectivity ≥ 2), and dual-scenario pumped-storage power flow security, while resilience is quantified by recovery-distance contribution and capacity-weighted betweenness centrality. The construction process follows four fundamental principles, which are systematically incorporated into the subsequent model:

- a. Bidirectional power-carrying criterion: The grid structure must satisfy transmission capacity constraints under both pumped-storage power generation and pumping operation modes, ensuring bidirectional power flow security and balance.
- b. N-1 connectivity criterion: At least two edge-disjoint paths must exist between core-load buses and pumped-storage black-start power sources, ensuring a minimum edge connectivity of two to maintain power supply continuity during single component failures.
- c. Power supply assurance criterion: It must cover all critical power buses, core-load buses, and black-start buses (e.g., pumped-storage plants) to ensure physical connectivity of essential services under extreme conditions.
- d. Economic-resilience synergy principle: Under strict compliance with the aforementioned safety hard constraints, grid planning should balance construction costs with resilience benefits, offering diversified solutions ranging from economic priority to safety priority.

2.2. Technical Characteristics and Recovery Value of Pumped-Storage Plants

As a strategic resource for enhancing the safety and resilience of power systems, pumped-storage plants feature bidirectional power flow (generation and pumping), minute-level response capabilities, and black-start functionality [28]. For fixed-speed pumped-storage units, the pumping power P_{pump} is typically represented at the rated power P_N [29], whereas the generation power P_{gen} is allowed to vary within a practical range of 50–100% of P_N [30].

$$P_{pump} = P_N \quad (1)$$

$$0.5P_N \leq P_{gen} \leq P_N \quad (2)$$

This study integrates pumped storage into the black-start power supply pool for connectivity and N-1 connectivity verification. Typically connected to critical grid sections, these facilities exhibit high betweenness centrality at grid buses. Their strategic location enables efficient power distribution during system recovery. This hub value is quantified through recovery-distance contribution and capacity-weighted betweenness centrality, thereby incorporating the technical characteristics of pumped storage into the resilience objective and mapping them to system-level resilience gains.

3. Quantitative Index System for Grid Resilience Considering Pumped-Storage Characteristics

3.1. Quantitative Framework for Resilience Recovery

Grid resilience emphasizes maintaining critical services under extreme disturbances, differing from traditional reliability, with a focus on post-disturbance recovery speed and continuity of system functionality [31,32]. Existing studies have shown that power system resilience assessment usually needs to consider both structural and functional changes, since a single performance perspective is insufficient to fully characterize the system state after disturbances [33]. Similarly, comprehensive resilience assessment frameworks also emphasize constructing integrated indicators from multiple dimensions to improve the overall characterization of system states [34]. This study focuses on post-disaster scenarios, quantifying the backbone grid's power supply reconstruction capacity for core loads supported by pumped-storage black-start systems. This capacity depends on two factors: first, the electrical accessibility of recovery paths, referring to the electrical distance between core loads and black-start power sources; second, the hubness of pumped-storage networks, indicating their central role in the grid topology. By coupling recovery-distance contribution with capacity-weighted betweenness centrality, this paper constructs a system resilience mismatch index (F_2), where lower values indicate better system resilience.

3.2. Contribution of Distance to Restoration

The recovery-distance contribution $F_{2,\text{dist}}$ quantifies the electrical cost required to restore power from a pumped-storage (black-start) bus to a core-load bus after a disturbance. A lower value indicates shorter recovery paths and faster restoration speeds. Existing complex-network resilience studies have shown that shortest paths and their aggregated forms can effectively characterize network accessibility and recovery capability [34]. The calculation formula is:

$$F_{2,\text{dist}} = \frac{1}{T_{\text{norm}}} \cdot \frac{1}{|V_{\text{core}}|} \sum_{v \in V_{\text{core}}} \min_{u \in V_{\text{psb}}} d_w(v, u) \quad (3)$$

where $|V_{\text{core}}|$ is the number of core-load buses; V_{psb} is the set of black-start buses (i.e., pumped-storage buses); T_{norm} is the normalization factor; and $d_w(v, u)$ is the weighted shortest path distance from a bus v to a black-start bus u .

The weighted shortest path distance $d_w(v, u)$ is based on line reactance and includes a capacity adjustment factor:

$$d_w(v, u) = \min_{p \in P(v, u)} \sum_{i \in p} (x_i \times 100 \times k_i) \quad (4)$$

where x_i is the reactance of line i on the path p , and the multiplier 100 is introduced as a scaling factor to keep the resulting value within a reasonable optimization range.

The capacity adjustment factor k_i is defined as:

$$k_i = \frac{1}{1 + S_i/1000} \times \xi_i \quad (5)$$

where S_i denotes the rated capacity of line i , reflecting the engineering principle that a higher-capacity line generally provides stronger power-transmission capability per unit

reactance. A special discount factor ξ_i is applied when a transmission line is directly connected to a pumped-storage bus, reflecting the greater recovery value of pumped-storage black-start transmission channels due to their flexible and rapid response capability. Finally, the normalization factor T_{norm} is defined as the maximum weighted shortest-path distance from all pumped-storage buses in the original grid to the core-load buses, ensuring that the resulting value falls within the range (0, 1).

3.3. Pumped-Storage Hub Effect

The pumped-storage hub effect quantifies the criticality of pumped-storage buses in the network using capacity-weighted betweenness centrality, integrating both topological centrality and available power capacity [35]. Buses that lie on a larger number of shortest paths tend to act as hubs in power transmission, and their strategic locations provide enhanced support for post-disturbance recovery. Defined as:

$$BC_{psh} = \frac{\sum_{u \in V_{psh}} C_u \cdot BC(u)}{\sum_{u \in V_{psh}} C_u} \quad (6)$$

where C_u is the rated capacity (MW) of the u -th pumped-storage bus and $BC(u)$ is the betweenness centrality of the pumped-storage bus, calculated based on weighted distance:

$$BC(u) = \sum_{s \neq u \neq t} \frac{\sigma_{st}(u)}{\sigma_{st}} \quad (7)$$

where the metric consists of two components: the total number of shortest path routes σ_{st} between buses and the count of paths $\sigma_{st}(u)$ traversing specific buses. Topological centrality alone cannot capture actual support capacity. By capacity weighting, the system provides a comprehensive evaluation of pumped-storage buses' hub status and operational potential, effectively reflecting both the topological hub role and power support capability of pumped-storage facilities.

3.4. System Resilience Mismatch Index

The recovery distance factor and the pumped-storage hub effect factor are coupled through a product model. Similarly, comprehensive resilience assessment studies emphasize the coordinated integration of structural and functional indicators to improve the overall characterization of system-state changes [33,34]. Accordingly, the system resilience mismatch index is defined as:

$$F_2 = F_{2,dist} \times (1 - \alpha \cdot BC_{psh}) \quad (8)$$

where $\alpha = 0.3$ is the hub effect influence coefficient. A smaller value of F_2 indicates a stronger ability of the system to maintain power supply to core loads after disturbances and therefore reflects better resilience. The product form reflects that restoration distance represents the baseline structural difficulty of post-disturbance recovery, whereas the pumped-storage hub effect acts as a proportional correction term that captures the enhancement introduced by topological centrality and available support capacity.

The hub effect influence coefficient α is introduced to regulate the correction strength of the pumped-storage hub effect on the restoration distance. In this study, α is set to 0.3, which reflects the positive contribution of pumped-storage hub support to system recovery while avoiding over-amplification of a single topological attribute, thereby preserving engineering interpretability and numerical stability.

Since both $F_{2,dist}$ and BC_{psh} are determined by the network topology, line parameters, and pumped-storage capacities, they remain constant with respect to α under a fixed backbone-grid configuration. Therefore, the resilience mismatch index $F_2 = F_{2,dist} \times (1 - \alpha \cdot BC_{psh})$ is a linear function of α . The first-order derivative is given by $\frac{\partial F_2}{\partial \alpha} = -F_{2,dist} BC_{psh}$, indicating that increasing α enhances the reduction effect of the hub term

on resilience mismatch. Moreover, $\frac{\partial^2 F_2}{\partial \alpha^2} = 0$, showing that no nonlinear amplification or abrupt variation is introduced by α .

Additional comparisons among alternative surrogate formulations, together with the sensitivity analysis of α , are provided in Appendix A. These supplementary analyses are designed to examine the local stability and engineering interpretability of the proposed index under representative backbone configurations and typical disturbance scenarios, rather than to establish a strict global optimality proof for all parameter combinations.

4. Mathematical Model of the Backbone-Grid Structure

4.1. Decision Variables

Define the decision variable $y_i \in \{0, 1\}$, which indicates whether candidate line i is selected for the backbone grid. Let the set of candidate routes be denoted as E , and the decision for each route is:

$$y_i = \begin{cases} 1, & \text{if the route is selected for the backbone grid} \\ 0, & \text{if the route is not selected} \end{cases} \quad \forall i \in E \quad (9)$$

where the decision vector is defined as $\mathbf{Y} = [y_1, y_2, \dots, y_{|E|}]^T$. Within this framework, $L' = \{i \in E \mid y_i = 1\}$ represents the set of lines selected for the backbone network from the original candidate set E , and the resulting network topology is constructed accordingly.

4.2. Multi-Objective Function

The backbone network framework adopts a bi-objective optimization model to balance economic performance and resilience. The economic indicator is quantified using normalized impedance to represent construction costs and transmission losses, formulated as:

$$\min F_1(\mathbf{y}) = \frac{\sum_{i \in E} z_i \cdot y_i}{Z_{norm}} \quad (10)$$

where $z_i = \sqrt{r_i^2 + x_i^2}$ denotes the impedance modulus of line i , and Z_{norm} is the normalization factor set to three times the total impedance of the minimum spanning tree across the network, which ensures that the value of F_1 falls within a reasonable range and aligns with the target magnitude of resilience.

The resilience target is represented by the system resilience mismatch index, expressed as:

$$\min F_2(\mathbf{y}) = F_{2,dist}(\mathbf{y}) \times (1 - \alpha \cdot BC_{psh}(\mathbf{y})) \quad (11)$$

where $F_{2,dist}$ is the recovery-distance factor defined as the weighted electrical distance from a core-load bus to a black-start bus, BC_{psh} denotes the weighted centrality of the pumped-storage buses; and $\alpha = 0.3$ is the hub-effect coefficient controlling the discount intensity of the hub effect on resilience loss.

4.3. Constraints

To ensure the engineering feasibility of the backbone grid, the following three categories of hard constraints must be satisfied:

4.3.1. Connectivity Constraint

The backbone grid must form a connected graph, with all critical buses belonging to the same connected component.

$$V_{key} \subseteq V(C_1), |C| = 1 \quad (12)$$

where $V_{key} = V_{bs} \cup V_{core} \cup V_{imp}$ represents the set of all critical buses (black-start buses, core-load buses, and important power source buses), $V(C_1)$ denotes the set of buses in the largest connected component (or any connected component) of the network $G(L')$

formed by the selected transmission lines, and $|C|$ represents the total number of connected components in the network $G(L')$.

4.3.2. N-1 Connectivity Constraint

This constraint ensures power supply continuity under single-component failures. It requires at least two edge-disjoint paths between core-load buses and pumped-storage buses, with the edge connectivity satisfying:

$$\lambda_{G(L')}(v, u) \geq K, \forall v \in V_{\text{core}}, \forall u \in V_{\text{psh}} \quad (13)$$

where $\lambda_{G(L')}(v, u)$ denotes the edge connectivity between bus v and bus u in graph $G(L')$, and $K = 2$ is imposed to ensure that single-line faults do not affect the power supply to core-load buses.

4.3.3. Power Flow Safety Constraints

The power flow safety constraints require the backbone grid to satisfy safe operating conditions under both power generation and pumping scenarios of pumped-storage units. These constraints are evaluated using a DC power flow model and transformed into continuous risk indices to facilitate evolutionary search, as detailed below.

In system operation analysis, the DC power flow model is adopted to approximate active-power flows on transmission lines. Owing to its simple formulation and computational efficiency, DC power flow is widely used in planning and optimization studies of high-voltage transmission systems [36]; however, it does not explicitly model voltage magnitudes, reactive power, or network losses, and therefore the results are more suitable for planning-stage backbone-grid screening rather than detailed operational validation [37]. The relationship between bus active power injection and bus voltage phase angles can be expressed as:

$$\mathbf{P} = \mathbf{B}\boldsymbol{\theta} \quad (14)$$

where $\mathbf{P}$ is the vector of active power injected at buses, $\mathbf{B}$ is the bus reactance matrix, and $\boldsymbol{\theta}$ is the vector of bus voltage phase angles. For any transmission line l connecting buses m and n , the active power flow is given by:

$$P_l = \mathbf{b}_l \cdot (\theta_m - \theta_n) \quad (15)$$

$$b_i = 1/x_i \quad (16)$$

where x_l is the reactance of line l .

To ensure system operational safety, line power flows must not exceed their transmission capacities, and bus phase angle differences must be controlled within allowable ranges:

$$|P_i| \leq \bar{P}_i, \forall i \in L' \quad (17)$$

$$|\theta_m - \theta_n| \leq \bar{\theta}, \forall \text{line } i \text{ connecting nodes } m, n \quad (18)$$

where $\bar{P}$ is the upper limit of active transmission capacity for line i , and $\bar{\theta}$ is the maximum allowable bus phase angle difference. In optimization problems, directly handling these inequality constraints often leads to a narrow feasible domain, which is not conducive to the efficient exploration of evolutionary algorithms. To this end, the hard constraints are transformed into continuous risk indices to quantify the degree of violation of safety constraints by candidate solutions.

The line overload risk is defined as:

$$R_{\text{line}} = \max_{i \in L'} \max \left(0, \frac{|P_i| - P_i^{\max}}{P_i^{\max}} \right) \quad (19)$$

The phase angle limit violation risk is defined as:

$$R_{\text{angle}} = \max_{(m,n) \in L} \max \left(0, \frac{|\theta_m - \theta_n| - \Delta\theta_{\max}}{\Delta\theta_{\max}} \right) \quad (20)$$

The total risk for a single operating scenario is the maximum of these two risks:

$$R_{\text{scenario}} = \max(R_{\text{line}}, R_{\text{angle}}) \quad (21)$$

Two typical operating scenarios are considered to comprehensively assess system safety. In the generation scenario, the load is at 100% of its base value, and pumped-storage units operate at 50–100% of their rated capacity for power generation. The bus power balance equation is:

$$\sum_{i \in G_{\text{conv}}} P_{i,\text{conv}}^{\text{gen}} + \sum_{u \in V_{\text{psh}}} P_{u,\text{psh}}^{\text{gen}} = \sum_{k \in L} P_{k,\text{load}}^{\text{base}} \times 1.0 \quad (22)$$

In the pumping scenario, the load is at 70% of its base value, and pumped-storage units operate at 100% of their rated capacity for pumping (with negative power). The bus power balance equation is:

$$\sum_{i \in G_{\text{conv}}} P_{i,\text{conv}}^{\text{pump}} - \sum_{u \in V_{\text{psh}}} P_{u,\text{psh}}^{\text{pump}} = \sum_{k \in L} P_{k,\text{load}}^{\text{base}} \times 0.7 \quad (23)$$

where G_{conv} represents the set of conventional generator buses, $P_{i,\text{conv}}^{\text{gen}}$ is the output power of conventional generator i , V_{psh} is the set of pumped-storage buses, $P_{u,\text{psh}}^{\text{pump}}$ denotes the power generation and pumping power of pumped-storage unit u , and $P_{k,\text{load}}^{\text{base}}$ is the base load at load bus k . The combined risk for the two scenarios is the maximum of the total risks from each scenario:

$$R_{\text{total}} = \max(R_{\text{pump}}, R_{\text{gen}}) \quad (24)$$

The final feasible solution must strictly satisfy all safety constraints, i.e., the combined risk must be zero:

$$R_{\text{total}} = 0 \quad (25)$$

This risk index transforms hard constraints into continuous quantitative measures of constraint violation, providing more refined guidance for the multi-objective evolutionary algorithm and enabling the search process to adjust its direction according to risk levels while gradually approaching feasible high-quality solutions.

5. Integration of Three-Stage Framework and Graph-Theoretic Validation in TER-NSGA-II Algorithm

The construction of the backbone grid is a high-dimensional, discrete, and highly constrained multi-objective combinatorial optimization problem. Traditional mathematical programming methods are limited by combinatorial explosion and nonlinear constraints, making them inefficient for solving such problems. The standard NSGA-II algorithm performs well in general multi-objective optimization due to its elite retention, fast non-dominated sorting, and crowding distance mechanism [38]. However, it exhibits insufficient constraint handling capabilities when applied to engineering problems with strong constraints [39]. Firstly, the randomness of crossover and mutation often generates infeasible solutions that violate connectivity and N-1 connectivity constraints, and reliance on penalty functions severely reduces optimization efficiency. Secondly, it lacks precise quantification and evolutionary guidance for N-1 connectivity. Finally, it is prone to local optima and has difficulty maintaining population diversity, and its fixed elite-selection mechanism cannot meet the incremental needs of hierarchical constraints [40]. To address these issues, this paper proposes a three-stage evolutionary reverse-learning NSGA-II (TER-NSGA-II), which integrates a three-stage framework and graph-theoretic validation.

5.1. Three-Stage Iteration and Elite Selection Integrated Framework

The total number of iterations T_{max} is divided into three stages to match the engineering logic of “first connectivity, then redundancy, and finally safety”:

- a. Stage I (Connectivity Construction): Prioritize retaining individuals with the lowest constraint violation values, rapidly converging toward the feasible domain.
- b. Stage II (N-1 Connectivity Reinforcement): This stage balances N-1 connectivity and the system resilience mismatch index F_2 , giving individuals that satisfy N-1 connectivity a selection advantage. Those that do not satisfy the constraint are identified for reinforcement through the minimum cut set, and such edges are favored in genetic operations.
- c. Stage III (Convergence and Safety Optimization): Once the population has largely satisfied all hard constraints, the focus shifts to objective optimization and the retention of Pareto-front individuals.

The top 5 elite individuals from each generation directly enter the next generation, ensuring the stable transmission of high-quality solutions.

To enhance the interpretability of the staged search process and the transferability of parameter settings, the three parameters T_1 , T_2 , and $T_{reverse}$ are defined as the switching generations between Stage I and Stage II, between Stage II and Stage III, and the triggering interval of periodic reverse learning, respectively. These parameters jointly determine how search effort is allocated across the different stages. Considering their scale dependence on the total number of generations, an automatic configuration procedure based on irace is adopted in the case study section to determine their final values [41].

5.2. Hierarchical Constraint Screening Strategy

Rather than relying on the conventional “generate-repair” mode, hard constraints are transformed into dynamic selection pressures:

- a. Connectivity Handling: Quickly determine connectivity based on graph theory search, using “main-component anchoring + incremental component connection” repairs, with a preference for low-cost lines.
- b. N-1 Connectivity Assessment: Abstract the network as a unit-capacity undirected graph, with core loads as sources and pumped-storage as sinks, calculating edge connectivity using the max-flow min-cut theorem to identify key reinforcement edges that guide evolution.
- c. Power Flow Safety Check: In Stage III, individuals that meet connectivity and N-1 connectivity constraints undergo dual-scenario DC power flow calculations. A unified assessment is performed using dual-scenario (generation/pumping) risk indices R_{total} , with only individuals meeting safety standards retained. The risk indices provide a continuous search gradient, thereby avoiding the “cliff effect” caused by binary feasibility judgments.

5.3. Periodic Reverse Learning Mechanism

To further enhance the global search capability of TER-NSGA-II in discrete backbone-grid planning problems, a periodic reverse learning mechanism is introduced. The core idea is that, after several generations of evolution, part of the population may become clustered within a limited search region. If only conventional crossover and mutation operators are applied, the search process tends to remain confined to the current neighborhood, leading to reduced population diversity and an increased risk of premature convergence. Therefore, reverse learning is triggered every $T_{reverse}$ generations. By constructing reverse individuals with complementary line-selection patterns relative to the original

solutions, the search space is actively expanded, providing additional candidate solutions for subsequent environmental selection.

For any individual, its chromosome is defined as the binary decision vector $\mathbf{Y}$ in (9), where $y_i = 1$ indicates that candidate line i is selected into the backbone grid, and $y_i = 0$ otherwise. Reverse learning is implemented by performing a bitwise inversion on this binary structure. Given an original individual $\mathbf{Y}$, its reverse individual is defined as $\bar{\mathbf{Y}} = \mathbf{1} - \mathbf{Y}$, i.e., $\bar{y}_i = 1 - y_i$. This operation implies that the lines retained in the original solution are preferentially removed in the reversed solution, while the unselected lines are activated. As a result, the reverse individual is able to escape from the local structural pattern of the original solution and generate a new topology significantly different from the current population.

As illustrated in Figure 1, a simplified example is used to demonstrate the structural transformation of a reverse chromosome before and after connectivity repair. Since a feasible solution must ensure that key nodes are connected to the backbone grid, consider an individual with chromosome $\mathbf{Y}$, where selected candidate lines form a topology that satisfies the key-node connectivity requirement. After applying bitwise inversion, the reverse chromosome $\bar{\mathbf{Y}}$ is obtained. Due to the drastic change in line combinations, the reversed topology may violate connectivity constraints, potentially resulting in network disconnection or isolation of key nodes. Therefore, it cannot directly participate in environmental selection.

To address this issue, a connectivity repair procedure is first applied to the reversed individual. By adding necessary critical lines, the network is restored to a connected state. Let the repaired individual be denoted as $\mathbf{Y}^{rep}$, whose corresponding topology satisfies the connectivity requirement and can proceed to subsequent selection.

It should be noted that connectivity repair is only the first step for reintroducing reverse individuals into the candidate solution set. Based on the current evolutionary stage, additional constraint handling is performed. Specifically, in Stage II, further reinforcement of N-1 connectivity is conducted, while in Stage III, dual-scenario power flow security constraints are evaluated. In other words, reverse learning introduces complementary search directions rather than directly producing feasible solutions. Its effectiveness lies in the combined process of “reverse generation + constraint repair + environmental selection,” which introduces structurally diverse yet potentially feasible candidate solutions into the population.

From a procedural perspective, the periodic reverse learning mechanism can be summarized into five steps:

- Reverse learning is triggered when the current generation satisfies $t \bmod T_{\text{reverse}} = 0$;
- Each individual in the population undergoes bitwise inversion to generate reverse chromosomes;
- Connectivity repair is first applied, followed by stage-dependent constraint handling;
- The repaired reverse individuals are merged with the original population and offspring solutions to form an expanded candidate set;
- Non-dominated sorting, crowding distance calculation, and elite preservation are performed to select individuals for the next generation.

Therefore, periodic reverse learning is not a simple random perturbation, but a mechanism that constructs complementary search directions based on existing high-quality solutions. Through constraint repair and environmental selection, only promising solutions are retained. This mechanism expands the search space while maintaining solution quality, effectively reducing the risk of stagnation in local optima without disrupting the overall evolutionary framework of TER-NSGA-II.

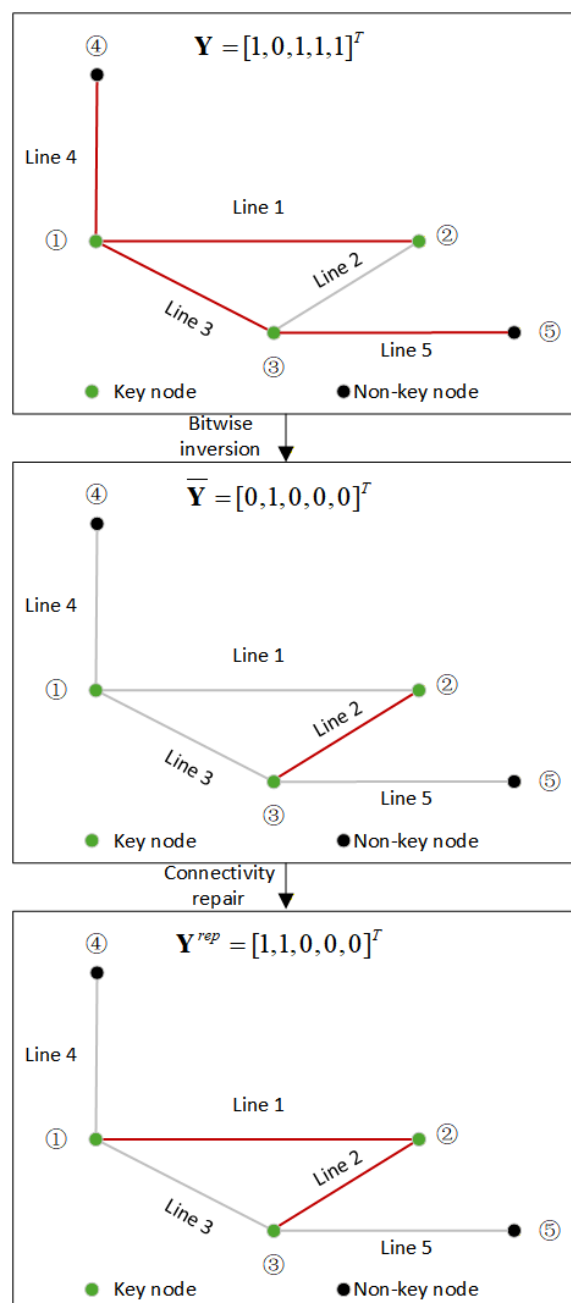


Figure 1. Illustration of reverse learning and connectivity repair. The circled numbers denote node indices, and the red lines indicate the active lines after connectivity repair.

5.4. Complete Solution Process of the TER-NSGA-II Algorithm

Based on the above improvement strategies, the complete solution process of the TER-NSGA-II algorithm is illustrated in Figure 2. The specific steps are as follows:

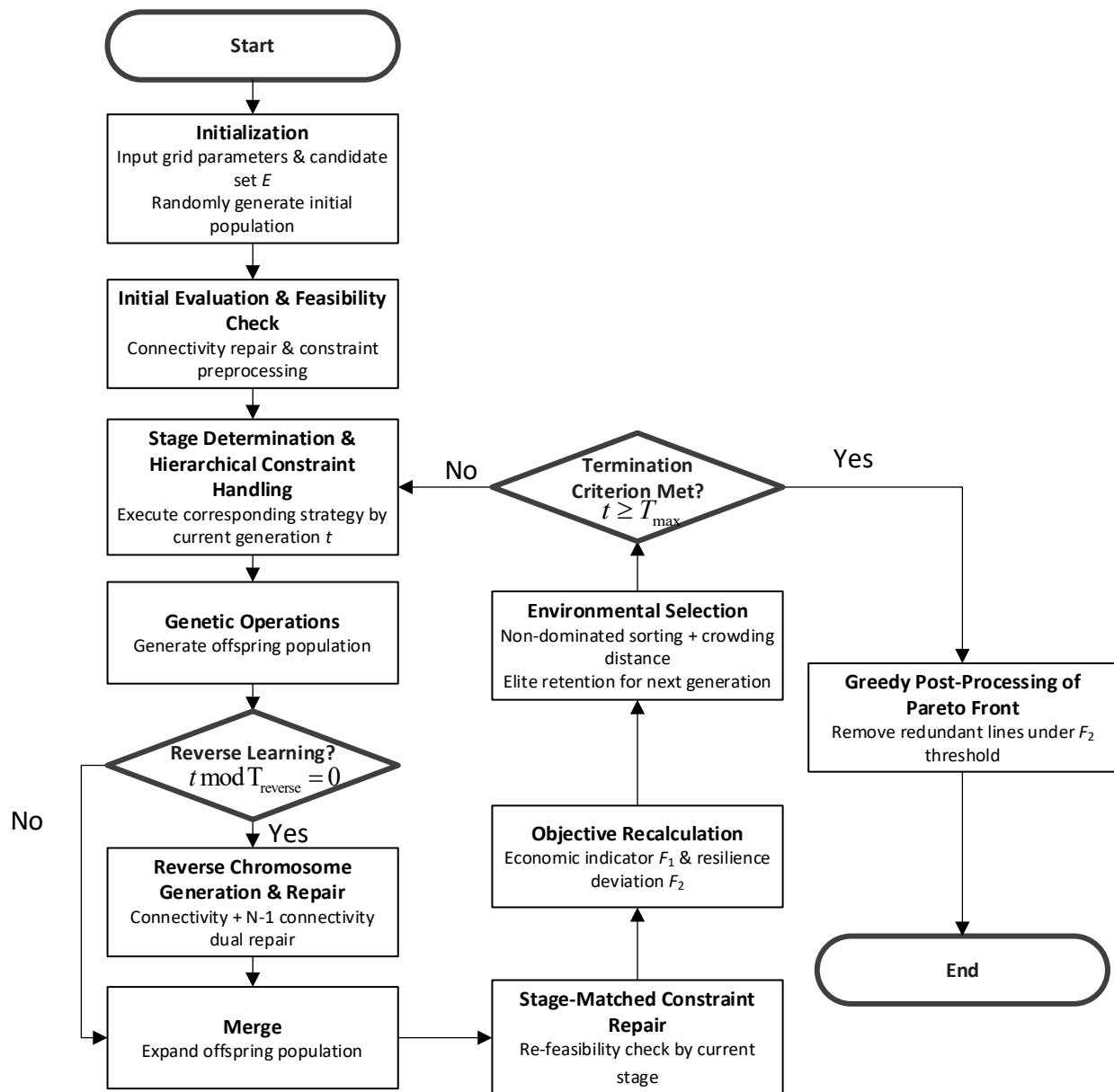


Figure 2. Overall solution process of the TER-NSGA-II algorithm.

- Input and Initialization: Input the original grid parameters and candidate line set E , initialize the genetic parameters, and randomly generate the initial binary population P_0 .
- Initial Evaluation: Calculate the economic objective F_1 and the system resilience mismatch index F_2 , and perform connectivity repair and constraint preprocessing to achieve initial feasibility.
- Stage Determination and Hierarchical Constraint Handling: Determine the current stage based on the current generation t and execute the corresponding hierarchical constraint strategy.
- Genetic Operations: Generate the offspring population Q_t through binary tournament selection, mask crossover, and bit-flip mutation.
- Trigger Reverse Learning: If $t \bmod T_{\text{reverse}} = 0$, generate reverse chromosomes and repair connectivity and N-1 connectivity, then merge and expand the offspring population to Q_t' ; otherwise, set $Q_t' = Q_t$.
- Constraint Repair and Objective Recalculation: Execute stage-matched constraint repair on Q_t' , and recalculate F_1 and F_2 .

- g. Environmental Selection: Merge Q_t' with the parent population P_t , perform non-dominated sorting and crowding distance calculation, and select the top M individuals to form the next generation P_{t+1} .
- h. Termination Check: If $t \geq T_{max}$, proceed to Step 9; otherwise, set $t = t + 1$ and return to Step 3.
- i. Post-processing and Output: Perform greedy post-processing on the Pareto front solution set. Under the condition that the change in the resilience index does not exceed a preset threshold, redundant lines are removed one by one while checking constraint satisfaction, and the final Pareto-optimal solution set for the backbone grid is then output.

5.5. Theoretical Performance Analysis of the Algorithm

The three-stage iterative framework activates constraints in layers, encouraging the population to gradually satisfy stricter engineering requirements and improving the feasible-domain search tendency. Let the proportion of individuals in the t -th generation satisfying the stage s constraint be denoted as $p_s(t)$. Stage I only requires connectivity constraints, so its feasible solution space $\mathcal{F}_1$ is much larger than that of Stage II ($\mathcal{F}_2$, which requires N-1 connectivity), and Stage III has the smallest feasible solution space $\mathcal{F}_3$ (which additionally requires power flow safety). Since the elite selection strategy preferentially retains individuals with low constraint violation values, and repair operations are guided in each generation, the following tendency can be expected:

$$p_1(t+1) \geq p_1(t), p_2(t+1) \geq p_2(t) \quad (26)$$

When switching between stages, the feasible solutions accumulated in the previous stage provide a high-quality initial population for the next stage, so the overall feasible-solution probability is expected to improve as the evolutionary process proceeds. In the IEEE 118-bus experiments, TER-NSGA-II achieved a 100.00% feasible-run rate across all independent runs, whereas NSGA-II achieved only 48.11%. This result is consistent with the above theoretical analysis.

The main computational cost of TER-NSGA-II comes from non-dominated sorting, constraint validation, and reverse learning. Let the population size be M , the maximum number of generations be T_{max} , and the number of candidate lines be H ($H = |E|$). The time complexity of non-dominated sorting is $O(M^2)$, and connectivity and N-1 connectivity validation for each pair of core-load and pumped-storage buses requires the Dinic algorithm, which has a complexity of $O(\min(V^{2/3}, H^{1/2})E)$. Given that in the IEEE 118-bus system, only 10 bus pairs need to be calculated, this part is actually much less time-consuming than non-dominated sorting. Therefore, the total time complexity is $O(T_{max}M^2 + T_{max}MH)$, which is higher than the standard NSGA-II's $O(T_{max}M^2)$, but due to the significant improvement in feasible solution rates, the algorithm avoids a large amount of redundant search. Under the same number of iterations, the probability of obtaining feasible high-quality solutions is therefore improved relative to that of traditional methods.

6. Case Studies and Numerical Validation

6.1. Test Systems and Experimental Settings

The IEEE 118-bus system is employed as the main test system to validate the proposed planning approach. The system configuration includes core generation buses (Buses 10, 69, 80, and 89), core-load buses (Buses 11, 16, 20, 29, and 50), and PSH units used as black-start resources (Buses 63 and 64, each with 240 MW). The planning model is subject to the N-1 connectivity constraint, which strictly requires an edge connectivity of $\lambda \geq 2$ between critical load centers and black-start units, together with dual-scenario steady-state security requirements.

To further examine the scalability of the proposed method, an additional IEEE 300-bus system is used for extended validation under the same modeling framework. This extended case is intended to assess the applicability of the proposed method to larger-scale networks and to examine whether the algorithm can maintain acceptable optimization performance as network size increases.

For TER-NSGA-II, the population size is set to 50 and the maximum number of generations is set to 200 in the IEEE 118-bus case. The stage-related parameters are determined through the irace-based tuning procedure described in Section 6.2, and the final configuration is set to $T_1 = 77$, $T_2 = 115$, and $T_{\text{reverse}} = 49$.

To evaluate the effectiveness of the proposed framework, NSGA-II and the more recent reference-point-based algorithm NSGA-III/NG are adopted as benchmark methods [42]. All compared algorithms use the same decision encoding, population size, termination criterion, network data, and constraint-evaluation criteria, so that the comparison is conducted under a unified problem definition and a consistent feasible-domain criterion.

6.2. Parameter Tuning and Configuration of TER-NSGA-II

To avoid relying entirely on empirical parameter settings, the key stage-related parameters of TER-NSGA-II are tuned using irace, an automatic algorithm-configuration procedure. Since T_1 , T_2 , and T_{reverse} are all scale-dependent with respect to the total number of generations G , they are first rewritten in normalized ratio form as $p_1 = T_1/G$, $p_2 = T_2/G$, and $p_r = T_{\text{reverse}}/G$. This ratio-based formulation allows the tuned configuration to better reflect the relative allocation of search effort across the three evolutionary stages.

During the tuning process, two aspects are considered simultaneously: the ability to generate feasible solutions under strong constraints and the overall bi-objective optimization performance. Several representative candidate configurations are screened and compared. As summarized in Table 1, the Baseline corresponds to the original empirical setting, whereas Candidate A, Candidate B, and Candidate C are representative configurations selected during the automatic tuning process. Although some candidates achieve lower scalarized tuning costs, they may still exhibit high zero-cost frequency, small feasible-solution-set size, or unstable performance across repeated runs. Therefore, the final parameter setting is selected not only according to the mean tuning cost, but also according to feasible-solution generation capability and run-to-run stability.

Table 1. Comparison of irace candidate parameter configurations.

Configuration	p_1	p_2	p_r	Mean Cost	Std	Feasible Count	Zero-Cost
Baseline	0.30	0.60	0.10	0.4077	0.1073	6.2	0
Candidate A	0.25	0.70	0.15	0.0344	0.0770	4.2	4
Candidate B	0.217	0.518	0.203	200,005	447,225	7.6	0
Candidate C	0.387	0.577	0.247	0.2232	0.1070	10.4	0

As shown in Table 1, Candidate A achieves the lowest mean cost, but it has a relatively high zero-cost frequency and a lower average feasible count, indicating possible solution-set degeneration or an overly small feasible solution set in some runs. Candidate B exhibits abnormally large mean cost and standard deviation, suggesting unstable performance. By contrast, Candidate C does not yield the lowest mean cost, but its zero-cost frequency is zero and its feasible count is the highest among the compared configurations. Therefore, Candidate C provides the best overall balance between feasible-solution acquisition, optimization performance, and robustness. Accordingly, in the formal experiments with $G = 200$, the final stage parameters are set to $T_1 = 77$, $T_2 = 115$, and $T_{\text{reverse}} = 49$. This configuration implies that the early search effort is primarily allocated to feasible connectivity construction, followed by N-1 connectivity reinforcement, and finally by

safety-constrained convergence and Pareto-front refinement. Overall, this parameter configuration is consistent with the hierarchical solution logic of “connectivity first, redundancy second, and safety last”.

6.3. IEEE 118-Bus System: Main Validation Results

The IEEE 118-bus system is used as the main validation case. All three algorithms, namely TER-NSGA-II, NSGA-II, and NSGA-III/NG, are independently executed 50 times under the same problem setting and unified constraint-evaluation criteria. To ensure fairness, all statistics reported below are summarized from feasible runs and their corresponding Pareto solution sets.

Table 2 reports the core-objective statistics, including the economic objective F_1 , the system resilience mismatch index F_2 , and the feasible-run rate. Here, feasible-run rate denotes the proportion of independent runs in which at least one feasible solution satisfying connectivity, N-1 connectivity, and dual-scenario power flow security constraints is obtained. This indicator is used to reflect the algorithm’s ability to stably enter the feasible domain under strong engineering constraints.

Table 2. Statistical comparison of core-objective indicators on the IEEE 118-bus system.

Algorithm	F_1 Mean	F_1 Std. Dev.	F_1 Min	F_2 Mean	F_2 Std. Dev.	F_2 Min	Feasible-Run Rate
TER-NSGA-II	0.2276	0.0119	0.1904	0.4738	0.0056	0.4594	100.00%
NSGA-II	0.3065	0.0233	0.2570	0.5088	0.0211	0.4839	48.11%
NSGA-III/NG	0.2845	0.0249	0.2337	0.5071	0.0078	0.4893	100.00%

As shown in Table 2, TER-NSGA-II achieves the best average performance in both F_1 and F_2 . Compared with NSGA-II, the mean value of F_1 is reduced by approximately 25.7%, while the mean value of F_2 is reduced by approximately 6.9%. Compared with NSGA-III/NG, the corresponding reductions are approximately 20.0% and 6.6%, respectively. These results indicate that, while maintaining strict feasibility, the proposed method can simultaneously obtain lower economic cost and better resilience performance.

In terms of result stability, TER-NSGA-II also exhibits smaller standard deviations for both F_1 and F_2 , indicating more stable optimization outcomes across repeated runs. More importantly, the feasible-run rate of TER-NSGA-II reaches 100.00%, whereas NSGA-II achieves only 48.11%. This suggests that, under multi-level hard constraints involving connectivity, N-1 connectivity, and dual-scenario power flow security, the staged constraint-evolution mechanism can guide the search process into the feasible domain more reliably and avoid runs that fail to produce any feasible solution.

To further analyze convergence behavior, a synthesized objective is defined as $f_{\text{composite}} = \frac{F_1 + F_2}{2}$. The corresponding convergence curves are shown in Figure 3. TER-NSGA-II enters a relatively competitive objective region at an earlier stage and maintains a more consistent downward trend throughout the evolutionary process. This indicates that the staged search strategy improves the effective search efficiency per iteration by successively focusing on connectivity construction, N-1 connectivity reinforcement, and safety-constrained convergence.

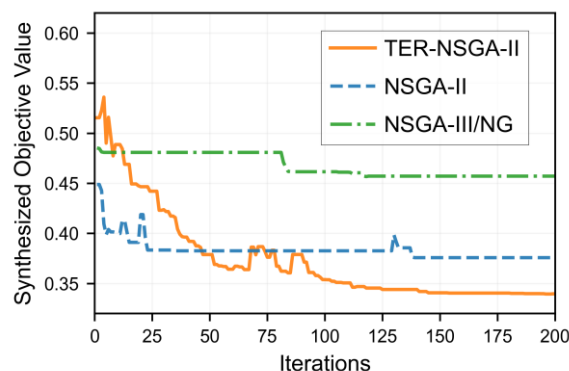


Figure 3. Convergence curves of the synthesized objective for different algorithms on the IEEE 118-bus system.

To evaluate the multi-objective solution-set quality, IGD+, HV, and Spread are adopted [43]. The corresponding results are shown in Table 3, and the representative Pareto fronts are shown in Figure 4.

Table 3. Multi-objective performance metrics on the IEEE 118-bus system.

Algorithm	IGD+	HV	Spread
TER-NSGA-II	0.0132	0.2632	1.0415
NSGA-II	0.0582	0.2012	1.0984
NSGA-III/NG	0.0124	0.0412	0.8869

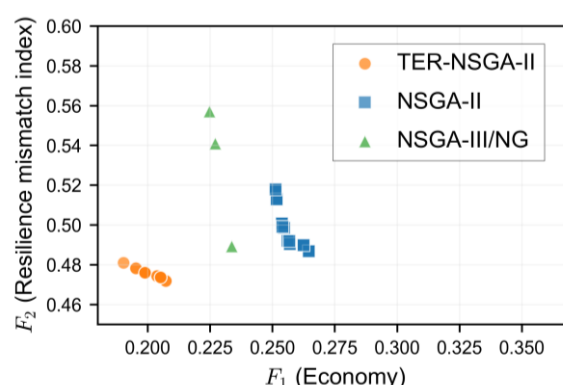


Figure 4. Comparison of representative Pareto fronts in the objective space on the IEEE 118-bus system.

According to Table 3 and Figure 4, TER-NSGA-II achieves the best HV value, indicating better coverage of the objective space and a larger effective non-dominated region. Although NSGA-III/NG attains an IGD+ value close to that of TER-NSGA-II, its HV value is substantially lower, suggesting that its solution set is relatively sparse and lacks sufficient coverage in the objective space. Therefore, although NSGA-III/NG obtains the smallest IGD+ value, TER-NSGA-II provides a more balanced overall Pareto-set performance in terms of convergence, coverage, and engineering feasibility.

6.4. Validation of the Capacity-Weighted Betweenness Centrality

To validate the mathematical formulation of the capacity-weighted betweenness centrality (BC_{psh}) proposed in this study, a comparative analysis was conducted using five independent Pareto solution sets. By evaluating three distinct weighting schemes—arithmetic mean, geometric mean, and unweighted average—the objective function F_2 was

recalculated to reconstruct the Pareto fronts for each approach. The multi-objective performance was then statistically evaluated using IGD+, Hyper-volume (HV), Spread, and the mean value of F_2 , as summarized in Table 4.

Table 4. Performance comparison of different aggregation forms for the pumped-storage hub effect.

Weighting Scheme	IGD+	HV	Spread	F_2 Mean
Arithmetic Mean (Proposed)	0.007524	0.108089	0.849734	0.474463
Geometric Mean	0.007632	0.107918	0.849441	0.474818
Unweighted Average	0.008262	0.107140	0.858419	0.476291

As shown in Table 4, the arithmetic mean (proposed) demonstrates the best or highly competitive performance in most metrics. Its HV value of 0.108089 is marginally higher than the 0.107918 yielded by the geometric mean and the 0.107140 from the unweighted average. Notably, the F_2 mean reaches its minimum value at 0.474463 under the proposed scheme, indicating a more precise characterization of the system resilience mismatch index.

Although the numerical improvements appear modest, a more detailed comparison reveals that, compared with the geometric mean, the arithmetic mean improves HV by approximately 0.16%, reduces the F_2 mean by approximately 0.07%, and reduces IGD+ by approximately 1.42%. Compared with the unweighted average, it improves HV by approximately 0.89%, reduces Spread by approximately 1.01%, reduces the F_2 mean by approximately 0.38%, and reduces IGD+ by approximately 8.94%. These disparities stem exclusively from the distinct weighting formulations, as the underlying solution sets remain identical. This confirms that the arithmetic mean provides a more rational integration of the capacity and topological contributions of pumped-storage nodes. It successfully avoids the over-compression of high-capacity hub effects inherent in the multiplicative nature of the geometric mean, while also rectifying the unweighted average's insensitivity to capacity disparities. Consequently, although the geometric mean yields a marginally smaller Spread value, the arithmetic-mean formulation achieves better IGD+, HV, and F_2 mean values, and therefore shows slightly better overall performance in this comparative post-processing test. It is adopted in this study as the aggregation form for quantifying the pumped-storage hub effect.

6.5. Analysis of Optimization Results and Engineering Schemes

Based on the Pareto optimal set generated by the TER-NSGA-II algorithm, three representative configuration candidates—Economic-oriented, Transitional, and Resilience-oriented schemes—are identified to meet diverse engineering requirements. Table 5 summarizes the quantitative indicators for these candidates, while their corresponding backbone network topologies are visualized in Figure 5a–c.

Table 5. Representative planning schemes under different economy-resilience preferences.

Configuration Type	F_1 (Economy)	F_2 (Resilience)	Selected Lines
Economic-oriented	0.190359	0.480958	54
Transitional scheme	0.218881	0.479010	67
Resilience-oriented	0.221911	0.459375	70

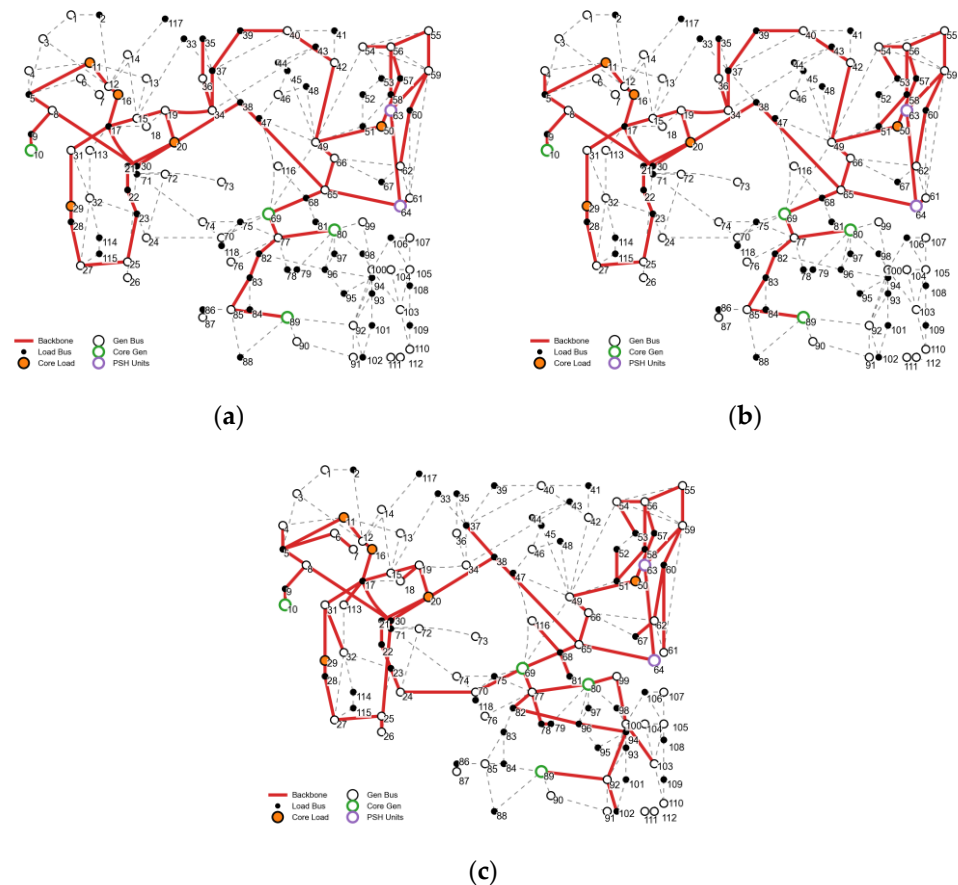


Figure 5. Topological configurations of the selected backbone-grid schemes: (a) Economic-oriented; (b) Transitional scheme; (c) Resilience-oriented. The numbers in the subfigures denote bus indices.

The Economic-oriented scheme features a minimal configuration of 54 transmission lines, indicating that it primarily satisfies the basic backbone connectivity required for system restoration, with a relatively compact structure. In contrast, the Resilience-oriented scheme incorporates 70 lines and achieves the lowest F_2 value by strengthening critical corridors and network redundancy to enhance system restoration support. The Transitional scheme, comprising 67 lines, lies between the other two schemes in terms of both line scale and objective values. Its structure preserves a certain level of economic efficiency while enhancing the supporting relationships among critical regions. These results indicate that different planning preferences lead to distinct backbone-grid structures, and the proposed method can provide differentiated planning schemes that balance economy and resilience under strict feasibility constraints.

6.6. Extended Validation on the IEEE 300-Bus System

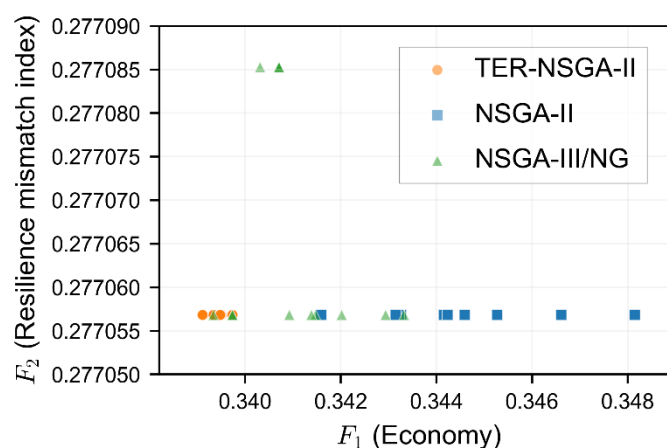
To further examine the applicability of the proposed method to larger-scale networks, extended validation is conducted on the IEEE 300-bus system. The experimental framework remains consistent with that of the IEEE 118-bus case, and all compared algorithms are independently executed 10 times. This part mainly focuses on the scalability of the proposed method, the stability of the optimization results, and the overall quality of the obtained Pareto solution sets. Table 6 summarizes the core-objective statistics on the IEEE 300-bus system.

Table 6. Statistical comparison of core-objective indicators on the IEEE 300-bus system.

Algorithm	F_1 Mean	F_1 Std. Dev.	F_1 Min	F_2 Mean	F_2 Std. Dev.	F_2 Min
TER-NSGA-II	0.339519	0.000216	0.339111	0.277057	0.000000	0.277057
NSGA-II	0.344422	0.001885	0.341591	0.277057	0.000000	0.277057
NSGA-III/NG	0.340238	0.000546	0.339337	0.277074	0.000015	0.277057

As shown in Table 6, all three algorithms are able to obtain feasible solutions on the IEEE 300-bus system, indicating that the problem remains computationally tractable as network scale increases. However, from the perspective of objective performance, TER-NSGA-II still achieves the lowest mean value of F_1 and the smallest fluctuation, indicating better optimization stability in the larger-scale case. The differences in F_2 are relatively small, suggesting that the numerical discrimination of the resilience objective becomes weaker in this test system. Nevertheless, the overall ranking pattern remains broadly consistent with that observed in the IEEE 118-bus case.

To further evaluate the Pareto-set quality, Table 7 reports IGD+, HV, and Spread on the IEEE 300-bus system, while Figure 6 presents the representative Pareto fronts.

**Figure 6.** Comparison of representative Pareto fronts in the objective space on the IEEE 300-bus system.**Table 7.** Multi-objective performance metrics on the IEEE 300-bus system.

Algorithm	IGD+ Mean	IGD+ Std	HV Mean	HV Std	Spread Mean	Spread Std
TER-NSGA-II	0.000040	0.000030	0.010866	0.000020	1.000000	0.000000
NSGA-II	0.002686	0.001492	0.010376	0.000179	1.000000	0.000000
NSGA-III/NG	0.000232	0.000147	0.010794	0.000052	0.848641	0.103905

These results show that TER-NSGA-II still maintains competitive Pareto-set quality in the larger-scale system. In particular, it achieves the best IGD+ and HV values, indicating that the obtained solution set remains close to the reference front and covers a relatively large objective-space region, although its Spread value is not the smallest among the compared algorithms. Therefore, the proposed method is not limited to a single medium-scale benchmark but can also be extended to a larger-scale network while maintaining acceptable optimization performance.

6.7. Computational Performance Analysis

In addition to solution-set quality and feasible-domain search capability, computational performance is also important for evaluating engineering applicability. Therefore, the average runtime and feasible-solution statistics of the compared algorithms on both the IEEE 118-bus and IEEE 300-bus systems are further analyzed, as summarized in Table 8.

Table 8. Computational performance of the compared algorithms on the IEEE 118-bus and IEEE 300-bus systems.

System	Algorithm	Runtime (s)	Feasible-Solution Count
IEEE 118-bus	TER-NSGA-II	1336.49 ± 99.93	24.42 ± 8.70
IEEE 118-bus	NSGA-II	254.35 ± 38.60	22.00 ± 25.07
IEEE 118-bus	NSGA-III/NG	219.26 ± 94.29	3.66 ± 1.56
IEEE 300-bus	TER-NSGA-II	814.62 ± 129.84	50.00 ± 0.00
IEEE 300-bus	NSGA-II	437.71 ± 49.94	50.00 ± 0.00
IEEE 300-bus	NSGA-III/NG	631.00 ± 83.76	1.60 ± 0.52

As shown in Table 8, TER-NSGA-II requires more runtime than the benchmark algorithms on both systems. This is expected, because the proposed method explicitly introduces staged constraint guidance and graph-theoretic edge-connectivity validation, which inevitably increase the computational burden. However, this additional cost is accompanied by better feasible-domain search behavior and higher-quality feasible solution sets, especially in the strongly constrained planning problem considered in this study.

From the perspective of feasible-solution count, TER-NSGA-II yields the largest feasible-solution set on the IEEE 118-bus system. On the IEEE 300-bus system, TER-NSGA-II and NSGA-II both stably obtain complete feasible solution sets, whereas NSGA-III/NG yields substantially fewer feasible solutions. It should be noted that this indicator reflects the size of the feasible solution set obtained within a run, which is conceptually different from the feasible-run rate reported for the IEEE 118-bus case.

Overall, the proposed algorithm incurs a higher computational cost, but this cost is exchanged for more reliable feasible-domain search and stronger engineering applicability under stringent connectivity and security constraints. Thus, TER-NSGA-II provides an acceptable trade-off between computational burden and solution quality for the backbone-grid planning problem studied here.

7. Conclusions

This paper proposes a bi-objective resilient backbone-grid planning framework to address the coordination between economic performance and structural resilience, explicitly incorporating hierarchical constraints and pumped-storage hub effects. A three-stage constrained evolutionary algorithm TER-NSGA-II is developed to optimize the system investment cost (F_1) alongside the system resilience mismatch index (F_2). The primary findings are summarized as follows:

- a. A bi-objective planning framework integrating economic efficiency with pumped-storage hub effects has been established. The resilience evaluation system combines recovery-distance contribution and capacity-weighted betweenness centrality, enabling an explicit assessment of black-start restoration potential. Supplementary analyses show that the multiplicative surrogate form provides clearer discrimination under severe perturbations than the additive alternative, while the sensitivity analysis of α confirms the local stability and engineering interpretability of the proposed resilience mismatch index.

- b. The proposed TER-NSGA-II algorithm introduces a rigorous three-stage constraint-handling framework. By implementing the maximum flow-minimum cut theorem to verify edge connectivity during the N-1 connectivity enhancement stage, this approach substitutes traditional heuristic approximations with rigorous hard constraints. This methodological shift significantly improves feasible-domain search efficiency and ensures the strict integrity of the solution set.
- c. The IEEE 118-bus main case and the IEEE 300-bus extended validation show that the proposed method can obtain Pareto solutions that balance economy and resilience under strict edge-connectivity requirements. Compared with the selected benchmark algorithms, TER-NSGA-II exhibits competitive overall performance in terms of solution-set quality, feasible-domain search stability, and scalability. The representative planning schemes further indicate that different redundancy levels can provide differentiated planning options under different economic and resilience preferences.

It should be noted that this study adopts a deterministic modeling framework and uses DC power flow to represent planning-stage operating conditions. Therefore, renewable-generation variability, load uncertainty, voltage constraints, reactive-power effects, and network losses are not explicitly modeled. Future work will extend the present framework by incorporating uncertainty modeling and more detailed operational constraints, as well as by exploring the coordinated planning of pumped storage with other flexible resources in modern power systems.

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Data Availability Statement: The source code supporting the findings of this study is publicly available at: <https://github.com/ding-jin-xiu/TER-NSGA-II-backbone-grid-planning> (accessed on 3 June 2026). The repository includes the implementation of TER-NSGA-II and runnable scripts for the main experimental workflow. The original power-system case data and additional supporting data are available from the corresponding author upon reasonable request.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

TER-NSGA-II	Three-stage Evolutionary Reverse learning NSGA-II
Std. Dev.	Standard Deviation

Appendix A. Rationality and Sensitivity Analysis of the Resilience Mismatch Index

To further validate the rationality of the proposed system resilience mismatch index, this appendix provides supplementary analysis from three aspects: alternative surrogate comparison, mechanism interpretation, and parameter sensitivity analysis.

It should be noted that the results in this appendix fall into two categories. The first category is based on post-processing comparisons conducted on a fixed set of representative backbone-grid solutions, aiming to illustrate the discriminative ability of different surrogate forms. The second category is based on sensitivity analysis performed on a fixed Pareto solution set, aiming to evaluate the local stability of parameter α .

Appendix A.1. Comparative Analysis of Alternative Surrogate Forms

To examine the rationality of the multiplicative formulation, four surrogate forms of the system resilience mismatch index are constructed and evaluated on the same set of representative backbone-grid solutions in the IEEE 118-bus system. These include the multiplicative form, additive form, distance-only form, and hub-only form.

Among them, the multiplicative form is the formulation adopted in this paper, while the additive form is used to test whether linear aggregation can replace multiplicative coupling. The distance-only form and hub-only form are used to examine whether relying solely on recovery distance or hub effect is sufficient for resilience characterization.

Since this analysis is conducted by recalculating surrogate indices on the same set of obtained representative solutions, the observed differences are attributed to the structure of the indices rather than differences in the optimization process.

The comparison is conducted on three representative backbone-grid schemes, corresponding to economic-oriented, transitional, and resilience-oriented configurations. Two scenarios are considered: the baseline scenario and a severe perturbation scenario. The baseline scenario represents the intact network condition, while the severe perturbation scenario corresponds to a critical-line-drop condition, where the most influential transmission line for system recovery is removed to simulate structural degradation under post-disaster conditions.

The relative degradation of different surrogate forms under the severe perturbation scenario compared to the baseline is summarized in Table A1.

Table A1. Relative degradation of different surrogate indices under severe perturbation.

Backbone Scheme	Multiplicative Form	Additive Form
Economic	10.64%	2.82%
Transitional	14.21%	4.83%
Resilience	14.20%	4.82%

As shown in Table A1, for all three representative backbone-grid schemes, the multiplicative form exhibits significantly larger degradation than the additive form under severe perturbation.

This indicates that the multiplicative formulation can more clearly amplify the coupled degradation between increased recovery distance and weakened hub support under high-impact disturbances. In contrast, the additive formulation tends to compress differences between schemes.

Therefore, although both formulations can reflect disturbance effects, the multiplicative form provides stronger discriminative capability for resilience differences and is more suitable as a system-level resilience mismatch index.

Appendix A.2. Mechanism Interpretation of the Index Formulation

The multiplicative formulation $F_2 = F_{2,\text{dist}} \times (1 - \alpha \cdot BC_{\text{psh}})$ captures the coupling relationship between restoration path accessibility and pumped-storage hub support. From an engineering perspective, the support capability of pumped-storage nodes cannot be realized independently of effective restoration paths. Therefore, the hub effect is more appropriately modeled as a proportional correction to recovery distance, rather than as a

simple linear additive component. This formulation is consistent with resilience assessment studies that emphasize the coordinated modeling of structural accessibility and functional support capacity, as well as with complex-network approaches that use weighted betweenness centrality to characterize critical-node support.

From a mathematical perspective, the formulation satisfies clear directional consistency. Taking the partial derivative with respect to the restoration-distance term yields $\partial F_2 / \partial F_{2,dist} = 1 - \alpha BC_{psh} > 0$ within the feasible range $0 \leq \alpha BC_{psh} < 1$, indicating that larger restoration distance leads to higher resilience mismatch. Taking the derivative with respect to the hub effect gives $\partial F_2 / \partial BC_{psh} = -\alpha F_{2,dist} < 0$, shows that stronger hub support reduces the resilience mismatch.

Further, taking the derivative with respect to α yields $\partial F_2 / \partial \alpha = -F_{2,dist} BC_{psh} < 0$. Since both $F_{2,dist}$ and BC_{psh} remain constant under a fixed backbone configuration, it follows that $\partial^2 F_2 / \partial \alpha^2 = 0$, indicating that F_2 varies linearly with α and that no nonlinear amplification or abrupt change is introduced. Therefore, the sensitivity of α can be effectively evaluated through finite-value statistical analysis.

In addition, the formulation exhibits consistent boundary behavior. When $BC_{psh} = 0$, F_2 reduces to $F_{2,dist}$, corresponding to a distance-dominated metric. As BC_{psh} increases, the factor $(1 - \alpha \cdot BC_{psh})$ imposes a bounded reduction on restoration cost. Therefore, the multiplicative form preserves the dominant role of recovery distance while incorporating positive correction from pumped-storage hub support, making it more consistent with the engineering mechanism of post-disaster restoration than simple additive formulations.

Appendix A.3. Sensitivity Analysis of Parameter α

Based on the above theoretical analysis, sensitivity analysis is further conducted on a fixed Pareto solution set for three representative values: $\alpha = 0.1, 0.3$, and 0.5 . Since both $F_{2,dist}$ and BC_{psh} remain constant under a fixed backbone-grid configuration, F_2 varies linearly with α . Therefore, this analysis focuses on whether variations in α lead to abnormal fluctuations in index values, changes in scenario ranking, or instability in evaluation results. It should be noted that the results in this section reflect local stability on the current solution set rather than a complete proof of global robustness.

The statistical results of the multiplicative form under baseline and severe perturbation scenarios are summarized in Table A2.

Table A2. Statistical results of F_2 under different α values.

α	Baseline	Severe Perturbation
0.1	0.5321 ± 0.0307	0.6327 ± 0.0591
0.3	0.4982 ± 0.0266	0.5994 ± 0.0660
0.5	0.4644 ± 0.0250	0.5660 ± 0.0735

As shown in Table A2, with increasing α , the value of F_2 decreases continuously under both scenarios, without abnormal jumps or discontinuities.

At the same time, the baseline scenario consistently yields lower values than the severe perturbation scenario, indicating that the relative ordering between different scenarios remains stable.

These results suggest that variations in α primarily affect the numerical scale of the index, rather than altering the relative comparison among scenarios.

From an engineering perspective, α controls the strength of hub-effect correction. Smaller α leads to a distance-dominated metric, while larger α enhances the contribution of pumped-storage hub support.

Since the observed variation is smooth and interpretable, $\alpha = 0.3$ provides a balanced configuration, capturing hub-effect benefits without overemphasizing a single topological attribute.

References

1. Yang, W.; Zhao, Z.; Pérez-Díaz, J.I.; Hunt, J.D.; Vagnoni, E.; Nøland, J.K.; Quaranta, E.; Wang, R.; Li, X.; Cheng, Y. Pumped Storage Hydropower Operation for Supporting Clean Energy Systems. *Nat. Rev. Clean Technol.* **2025**, *1*, 454–473. <https://doi.org/10.1038/s44359-025-00057-x>.
2. Shen, J.; Wang, Y.; Hao, T.; Cheng, C. Pumped-Storage Renovation for Grid-Scale, Long-Duration Energy Storage. *Nat. Rev. Electr. Eng.* **2025**, *2*, 79–80. <https://doi.org/10.1038/s44287-025-00143-z>.
3. Enasel, E.; Dumitrascu, G. Storage Solutions for Renewable Energy: A Review. *Energy Nexus* **2025**, *17*, 100391. <https://doi.org/10.1016/j.nexus.2025.100391>.
4. Chivunga, J.N.; Lin, Z.; Blanchard, R. Power Systems' Resilience: A Comprehensive Literature Review. *Energies* **2023**, *16*, 7256. <https://doi.org/10.3390/en16217256>.
5. Yao, F.; Chau, T.K.; Zhang, X.; Iu, H.H.C.; Fernando, T. An Integrated Transmission Expansion and Sectionalizing-Based Black Start Allocation of BESS Planning Strategy for Enhanced Power Grid Resilience. *IEEE Access* **2020**, *8*, 148968–148979. <https://doi.org/10.1109/ACCESS.2020.3014341>.
6. Elalfy, D.A.; Gouda, E.; Kotb, M.F.; Bureš, V.; Sedhom, B.E. Comprehensive Review of Energy Storage Systems Technologies, Objectives, Challenges, and Future Trends. *Energy Strategy Rev.* **2024**, *54*, 101482. <https://doi.org/10.1016/j.esr.2024.101482>.
7. Paul, S.; Poudyal, A.; Poudel, S.; Dubey, A.; Wang, Z. Resilience Assessment and Planning in Power Distribution Systems: Past and Future Considerations. *Renew. Sustain. Energy Rev.* **2024**, *189*, 113991. <https://doi.org/10.1016/j.rser.2023.113991>.
8. Chen, X.; Liu, J.; Li, P.; Ren, J.; Zhang, D.; Zhou, X. A Multi-Stage Resilience Enhancement Method for Distribution Networks Employing Transportation and Hydrogen Energy Systems. *Sustainability* **2025**, *17*, 8691. <https://doi.org/10.3390/su17198691>.
9. Girma, W.; Awoke, A.G.; Hailu, H.; Melesse, A.M. Multi-Criteria Decision Analysis for Optimal Placement of Pumped Hydro-Energy Storages: The Case of Existing Hydropower Reservoirs of Ethiopia. *Appl. Water Sci.* **2025**, *15*, 9. <https://doi.org/10.1007/s13201-025-02524-z>.
10. Ji, L.; Jiang, Y.; Ji, S.; Chen, K. An Integrated Optimization Framework Unlocks Energy Storage Economic Value in Renewable Energy Bases through Planning Operation Coordination. *Discov. Appl. Sci.* **2025**, *8*, 59. <https://doi.org/10.1007/s42452-025-08077-9>.
11. Wang, C.; Ju, P.; Wu, F.; Pan, X.; Wang, Z. A Systematic Review on Power System Resilience from the Perspective of Generation, Network, and Load. *Renew. Sustain. Energy Rev.* **2022**, *167*, 112567. <https://doi.org/10.1016/j.rser.2022.112567>.
12. Shao, Y.; Gong, Y.; Wan, X.; Huang, X.; Luo, S.; Cao, Y.; Zhang, T. Quantum-Enabled Topological Optimization of Distributed Energy Storage for Resilient Black-Start Operations. *Sci. Rep.* **2025**, *15*, 18034. <https://doi.org/10.1038/s41598-025-02286-3>.
13. Liu, Z.; Wang, B.; Chen, Y.; Chen, Y.; Jiayang, L.; Zhang, Q.; Yuan, N.; Lu, Q.; Zhu, L.; Lin, Y. Optimal Configuration Strategy of Energy Storage for Enhancing the Comprehensive Resilience and Power Quality of Distribution Networks. *Front. Energy Res.* **2024**, *12*, 1472486. <https://doi.org/10.3389/fenrg.2024.1472486>.
14. Bhowmik, A.R.; Chakraborty, A.K.; Mukherjee, S. Impact of Sensitivity Index Based Transmission Network Topology Control on Contingency Constraints. In Proceedings of the 2019 20th International Conference on Intelligent System Application to Power Systems (ISAP), New Delhi, India, 10–13 December 2019; pp. 1–7. <https://doi.org/10.1109/ISAP48318.2019.9065962>.
15. Poyrazoglu, G.; Oh, H. Optimal Topology Control with Physical Power Flow Constraints and N-1 Contingency Criterion. In Proceedings of the 2015 IEEE Power & Energy Society General Meeting, Denver, CO, USA, 26–30 July 2015; p. 1. <https://doi.org/10.1109/PESGM.2015.7286191>.
16. Matthews, L.R.; Gounaris, C.E.; Kevrekidis, I.G. Designing Networks with Resiliency to Edge Failures Using Two-Stage Robust Optimization. *Eur. J. Oper. Res.* **2019**, *279*, 704–720. <https://doi.org/10.1016/j.ejor.2019.06.021>.
17. Zhao, H.; Xu, Y. Site Selection Evaluation of Pumped Storage Power Station Based on Multi-Energy Complementary Perspective: A Case Study in China. *Energies* **2025**, *18*, 3549. <https://doi.org/10.3390/en18133549>.
18. Chen, X.; Chen, C.; Tian, G.; Yang, Y.; Zhao, Y. Comprehensive Evaluation Research of Hybrid Energy Systems Driven by Renewable Energy Based on Fuzzy Multi-Criteria Decision-Making. *Front. Energy Res.* **2023**, *11*, 1294391. <https://doi.org/10.3389/fenrg.2023.1294391>.
19. Wu, Y.; Zhang, T.; Xu, C.; Zhang, X.; Ke, Y.; Chu, H.; Xu, R. Location Selection of Seawater Pumped Hydro Storage Station in China Based on Multi-Attribute Decision Making. *Renew. Energy* **2019**, *139*, 410–425. <https://doi.org/10.1016/j.renene.2019.02.091>.

20. Guo, Y.; Xiang, Y.; Tan, Z.; Zhang, H.; Li, J.; Hu, Z.; Liu, F.; Liu, J. Competitive Evaluation and Multi-Stage Planning of Park Integrated Energy Systems. *Sci. Rep.* **2025**, *15*, 8660. <https://doi.org/10.1038/s41598-025-92431-9>.
21. Gao, J.; Men, H.; Guo, F.; Liu, H.; Li, X.; Huang, X. A Multi-Criteria Decision-Making Framework for Compressed Air Energy Storage Power Site Selection Based on the Probabilistic Language Term Sets and Regret Theory. *J. Energy Storage* **2021**, *37*, 102473. <https://doi.org/10.1016/j.est.2021.102473>.
22. Alhazmi, M.; Li, P. Advancing Resilient Power Systems through Hierarchical Restoration with Renewable Resources. *Sci. Rep.* **2025**, *15*, 29755. <https://doi.org/10.1038/s41598-025-14992-z>.
23. Moreno, R.; Street, A.; Arroyo, J.M.; Mancarella, P. Planning Low-Carbon Electricity Systems under Uncertainty Considering Operational Flexibility and Smart Grid Technologies. *Philos. Trans. R. Soc. A Math. Phys. Eng. Sci.* **2017**, *375*, 20160305. <https://doi.org/10.1098/rsta.2016.0305>.
24. Fu, Y.; Lin, H.; Ma, C.; Sun, B.; Li, H.; Sun, Q.; Wennersten, R. Effects of Uncertainties on the Capacity and Operation of an Integrated Energy System. *Sustain. Energy Technol. Assess.* **2021**, *48*, 101625. <https://doi.org/10.1016/j.seta.2021.101625>.
25. Wang, H.; Ge, J.; Zhao, Y.; Gu, J.; Li, C.; Liu, C. Resilience enhancement strategies for distribution networks considering the coordination of 5G base stations and multiple flexible resources. *Sci. Rep.* **2026**, *16*, 5481. <https://doi.org/10.1038/s41598-026-35188-z>.
26. Sun, S.; Li, N.; Zhang, L.; Zhao, D.; Di, L.; Feng, L. Integrated data-driven topology reconstruction and risk-aware reconfiguration for resilient power distribution systems under incomplete observability. *Sci. Rep.* **2025**, *15*, 30661. <https://doi.org/10.1038/s41598-025-15440-8>.
27. Asim, A.M.; Awad, A.S.A.; Attia, M.A. Integrated optimization of energy storage and green hydrogen systems for resilient and sustainable future power grids. *Sci. Rep.* **2025**, *15*, 25656. <https://doi.org/10.1038/s41598-025-09408-x>.
28. Yang, F.; Wang, G.; An, S.; Su, H.; Zhang, Q.; Chen, J. Real-time control method and performance evaluation of pumped storage system based on adaptive moving average channel strategy. *Front. Energy Res.* **2025**, *13*, 1711460. <https://doi.org/10.3389/fenrg.2025.1711460>.
29. Joseph, A.; Desingu, K.; Semwal, R.R.; Chelliah, T.R.; Khare, D. Dynamic performance of pumping mode of 250 MW variable speed hydro-generating unit subjected to power and control circuit faults. *IEEE Trans. Energy Convers.* **2018**, *33*, 430–441. <https://doi.org/10.1109/TEC.2017.2739132>.
30. Luan, Y.H.; Zhang, Y.Z.; Zhuang, K.X.; Ding, L.J.; Sun, J.J. Research on power characteristics of full-power variable speed pumped storage units based on hydraulic-mechanical-electrical modeling. *Trans. China Electrotech. Soc.* **2025**, *40*, 730–743. <https://doi.org/10.19595/j.cnki.1000-6753.tces.240075>.
31. Wang, Y.; Chen, C.; Wang, J.; Baldick, R. Research on resilience of power systems under natural disasters—A review. *IEEE Trans. Power Syst.* **2016**, *31*, 1604–1613. <https://doi.org/10.1109/TPWRS.2015.2429656>.
32. Rouholamini, M.; Wang, C.; Magableh, S.; Wang, X. Resiliency of Electric Power Distribution Networks: A Review. *J. Infrastruct. Preserv. Resil.* **2025**, *6*, 39. <https://doi.org/10.1186/s43065-025-00154-y>.
33. Zhou, D.; Pan, X.; Sun, X.; Hu, F. Resilience assessment framework for high-penetration renewable energy power system. *Sustainability* **2025**, *17*, 2058. <https://doi.org/10.3390/su17052058>.
34. Zang, D.; Wei, C.; Zhao, J.; Zhang, J.; Tang, K.; Zhu, H. Predictive resilience assessment featuring diffusion reconstruction for road networks under rainfall disturbances. *Eng. Appl. Artif. Intell.* **2024**, *138*, 109317. <https://doi.org/10.1016/j.engappai.2024.109317>.
35. Zhang, C.; Wang, Y.; Zheng, T.; Zhang, K. Complex network theory-based optimization for enhancing resilience of large-scale multi-energy system. *Appl. Energy* **2024**, *370*, 123593. <https://doi.org/10.1016/j.apenergy.2024.123593>.
36. Stott, B.; Jardim, J.; Alsac, O. DC Power Flow Revisited. *IEEE Trans. Power Syst.* **2009**, *24*, 1290–1300. <https://doi.org/10.1109/TPWRS.2009.2021235>.
37. Yang, Z.; Zhong, H.; Xia, Q.; Kang, C. A Novel Network Model for Optimal Power Flow with Reactive Power and Network Losses. *Electr. Power Syst. Res.* **2017**, *144*, 63–71. <https://doi.org/10.1016/j.epsr.2016.11.009>.
38. Deb, K.; Pratap, A.; Agarwal, S.; Meyarivan, T. A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Trans. Evol. Comput.* **2002**, *6*, 182–197. <https://doi.org/10.1109/4235.996017>.
39. Tanabe, R.; Oyama, A. A Note on Constrained Multi-Objective Optimization Benchmark Problems. In Proceedings of the 2017 IEEE Congress on Evolutionary Computation (CEC), Donostia, Spain, 5–8 June 2017; pp. 1127–1134. <https://doi.org/10.1109/CEC.2017.7969433>.
40. Xiang, Z.H.; Wang, B.Z.; Xue, J.Z.; Shan, S. Research on the Practical Application of NSGA-II Algorithm for Multi-Objective Optimization. *Adv. Appl. Math.* **2023**, *12*, 4195–4207. <https://doi.org/10.12677/AAM.2023.1210413>.

41. López-Ibáñez, M.; Dubois-Lacoste, J.; Pérez Cáceres, L.; Birattari, M.; Stützle, T. The irace package: Iterated racing for automatic algorithm configuration. *Oper. Res. Perspect.* **2016**, *3*, 43–58. <https://doi.org/10.1016/j.orp.2016.09.002>.
42. Liu, Y.J. New Search Strategy for Multi-Objective Evolutionary Algorithm. *Heliyon* **2024**, *10*, e40917. <https://doi.org/10.1016/j.heliyon.2024.e40917>.
43. Ishibuchi, H.; Masuda, H.; Nojima, Y. Sensitivity of Performance Evaluation Results by Inverted Generational Distance to Reference Points. In Proceedings of the 2016 IEEE Congress on Evolutionary Computation (CEC), Vancouver, BC, Canada, 24–29 July 2016; pp. 1107–1114. <https://doi.org/10.1109/CEC.2016.7743912>.

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